

PCSK9 (Proprotein convertase subtilisin/kexin type 9) inhibitors: past, present, and the future

Yuichi J. Shimada, Christopher P. Cannon

Prevention

PCSK9 (Proprotein convertase subtilisin/kexin type 9) inhibitors: past, present, and the future

Yuichi J. Shimada^{1,2*} and Christopher P. Cannon^{1,2}

¹Harvard Clinical Research Institute, Boston, MA, USA; and ²Cardiovascular Division, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA

Received 29 January 2015; revised 20 April 2015; accepted 22 April 2015; online publish-ahead-of-print 13 May 2015

Reduction in low-density lipoprotein cholesterol (LDL-C), mainly with statins, has decreased the risk of cardiovascular events over the last few decades. However, there are several patient populations that warrant further decrease in LDL-C by additional cholesterol-lowering therapy other than statins. Proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors are a new class of drugs that have been shown to further decrease LDL-C by 50–70% when administered as a monotherapy or on a background therapy with statins. Proprotein convertase subtilisin/kexin type 9 inhibitors are also an excellent example of drug development in which discovery of gene mutations and its clinical effects have rapidly progressed into successful preclinical and clinical studies with multiple Phases 1–3 clinical trials completed or ongoing to date. This review summarizes the rapid evolution of the drug from genetic discovery to identification of targets for the drugs, to animal and human testing, and to large clinical outcomes trials, followed by discussion on foreseeable challenges of PCSK9 inhibitors.

Keywords

Proprotein convertase subtilisin/kexin type 9 inhibitors • Low-density lipoprotein cholesterol • Hypercholesterolaemia • Cardiovascular disease

Introduction

Cardiovascular disease accounts for the largest proportion in terms of the cause of death in advanced countries.¹ Low-density lipoprotein cholesterol (LDL-C) has been identified as an independent and most potent modifiable risk factor for developing cardiovascular disease.^{2,3} The use of statins over past 20 years to reduce LDL-C has successfully decreased the risk of cardiovascular events,^{4,5} reinforcing the role of lowering LDL-C as a major means for diminishing cardiovascular morbidity and mortality.

However, despite statin therapy, there still exists an unmet need for an additional/alternative LDL-C-lowering therapy for further risk reduction. For example, some patients, especially with familial hypercholesterolaemia (FH), cannot achieve the goal LDL-C concentrations even with the maximal dose of highly potent statins.⁶ Moreover, some patients cannot tolerate statins at all or can only take a small dose of statins due to various reasons such as myalgia and rhabdomyolysis. Proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors are a new class of drugs that appear promising as a novel agent for LDL-C reduction after being tested in several clinical trials. In this review, we will summarize the history and development as well as completed and ongoing clinical trials, and discuss some of the foreseeable challenges of PCSK9 inhibitors.

Genetics, discovery, and link to clinical outcomes

Genetics and molecular structure of proprotein convertase subtilisin/kexin type 9

Proprotein convertase subtilisin/kexin type 9 gene has the length of 3617 bp and expands over 12 exons encoding 692 amino acids for PCSK9.^{7,8} Expression of PCSK9 is particularly abundant in the liver, gastrointestinal system, kidney, and nervous system.^{8,9} Proprotein convertase subtilisin/kexin type 9 molecule consists of a signal sequence, a prodomain, a catalytic domain, and a C-terminal domain that contains abundant cysteine and histidine.^{10,11}

Discovery of clinical importance of proprotein convertase subtilisin/kexin type 9 gene mutations

The clinical effects of PCSK9 were first recognized when a mutation in the gene was detected in French families in 2003.¹² Subsequently, PCSK9 gene became the third gene after LDL receptor (LDL-R) gene

*Corresponding author. Tel: +1 617 307 5457, Fax: +1 617 307 5600, Email: yshimada@partners.org

Published on behalf of the European Society of Cardiology. All rights reserved. © The Author 2015. For permissions please email: journals.permissions@oup.com.

and apolipoprotein B gene that was associated with autosomal-dominant form of FH, which is caused by a gain-of-function mutation in PCSK9 gene. Elevated LDL-C concentration, in turn, results in excessive risk of cardiovascular disease in these patients.^{12–16} Mutations in PCSK9 gene are reported to be responsible for ~10–25% of autosomal-dominant form of FH cases without mutations in LDL-R or apolipoprotein B.^{13,15}

Clinical effects of loss-of-function mutation in proprotein convertase subtilisin/kexin type 9

In contrast to the gain-of-function mutations, mutations that cause loss of function of PCSK9 have been reported to decrease in LDL-C concentrations in various patient populations. The Dallas Heart Study reported that individuals with nonsense mutation in PCSK9 gene had 28% lower LDL-C concentrations than general population.¹⁷ The Atherosclerosis Risk In Communities study showed reduced concentrations of LDL-C by 15 and 28% in 9523 Caucasian and 3363 African-American subjects, respectively, by loss of function in one of the two PCSK9 genes.¹⁸ Data from the same cohort of patients have shown that heterozygous loss-of-function mutations in PCSK9 gene are associated with reduced rates of cardiovascular events. The hazard ratios for development of coronary heart disease during 15-year follow-up were 0.50 [95% confidence interval (CI) (0.32–0.79)] among Caucasians and 0.11 [95% CI (0.02–0.81)] in African-Americans.¹⁸

Similar findings were reported in two studies.^{19,20} In the Copenhagen Heart Study, the risk of developing coronary heart disease was reduced by 6–46% through reduction in LDL-C concentration by 11–15% in individuals with one loss-of-function mutation in PCSK9 gene.¹⁹ In the other study from Zimbabwe, the mean reduction in LDL-C concentration was 27% in African women.²⁰ Patients who are heterozygous for PCSK9 gene were also found to be protected against peripheral artery disease and have lower carotid intima media thickness.^{18,21,22} In healthy individuals with a nonsense mutation in each PCSK9 gene, LDL-C concentrations were as low as 14–16 mg/dL.^{23,24}

Inhibitors that target proprotein convertase subtilisin/kexin type 9

There are several PCSK9-directed therapies that are in various stages of development as summarized in Table 1. In this section, three

categories of PCSK9 inhibitors are discussed: monoclonal antibody, small interfering RNA (siRNA), and antisense oligonucleotides (ASOs).

Preclinical studies of monoclonal antibodies

Monoclonal antibodies are the most common method of PCSK9 inhibition since the first discovery in 2009.²⁵ These antibodies bind to the region on PCSK9 required for interaction with LDL-R, and inhibit interaction between PCSK9 and LDL-R. The first neutralizing anti-PCSK9 antibody increased LDL-R expressions on the hepatocytes and reduced LDL-C concentration by 30% in animals such as mice and primates.²⁵ Several other monoclonal antibodies have been discovered with dose dependent 20–50% reductions in the LDL-C concentrations in monkeys.^{26,27} It was reported in animals that the effects last for more than a few weeks.^{26,27} Based on these results, several monoclonal antibodies against PCSK9 have undergone Phases 1–3 clinical trials as discussed later.

Small interfering RNA

Another method of inhibiting the activities of PCSK9 is to administer a single-strand siRNA that binds the messenger RNA of PCSK9. These siRNAs can be administered intravenously in small lipid nanoparticles. Administration of siRNA against PCSK9 messenger RNA in rats showed silencing of more than a half of the messenger RNA, which led to reduction in the PCSK9 protein concentration in plasma, and in turn in LDL-C concentration by 30%.²⁸ A study in primates showed that the reduction in plasma PCSK9 concentrations by a single administration of siRNA was rapid, durable, and reversible.²⁸ In this study, LDL-C was reduced by 56–70% and the effects lasted for a few weeks. Promising results from a Phase 1 study were recently published regarding an siRNA; healthy volunteers who were given an siRNA had a 70% reduction in circulating PCSK9 concentrations and a 40% reduction in LDL-C from baseline.²⁹

Antisense oligonucleotides

Activities of messenger RNAs can be interfered with an ASO, which is a short nucleotide that binds to the messenger RNA.³⁰ A study on an ASO reported that messenger RNA was reduced by 92% and the LDL-C concentration by 32%.³¹ Another study in monkeys showed that circulating PCSK9 concentration was reduced by 85% and

Table 1 Selected proprotein convertase subtilisin/kexin type 9-directed therapies in development

Company	Drug	Agent	Indication	Phase
Sanofi/Regeneron	Alirocumab	Human monoclonal antibody	Hypercholesterolaemia	3 (published)
Amgen	Evolocumab	Human monoclonal antibody	Hypercholesterolaemia	3 (published)
Pfizer/Rinat	Bococizumab	Monoclonal antibody	Hypercholesterolaemia	3 (ongoing)
Novartis	LGT-209	Monoclonal antibody	Hypercholesterolaemia	2
Genentech	MPSK3169A, RG7652	Monoclonal antibody	Hypercholesterolaemia	2
Alnylam Pharmaceuticals/The Medicines Company	ALN-PCS02	siRNA oligonucleotide	Hypercholesterolaemia	1
Idera Pharmaceuticals	TBD	Antisense oligonucleotide	Hypercholesterolaemia	Preclinical

TBD, to be determined.
Modified from Urban et al.⁶⁹

LDL-C concentration by 50%.³² However, a Phase 1 clinical trial was terminated prematurely and there is currently no ongoing trial on ASO for PCSK9.³³

General mechanism of reduction in low-density lipoprotein cholesterol by proprotein convertase subtilisin/kexin type 9 inhibition

Physiological function of proprotein convertase subtilisin/kexin type 9

Secreted PCSK9 has two possible destinations: the first is to be secreted and immediately get bound to LDL-R in the liver, and the

second is to enter the systemic circulation. The PCSK9/LDL-R complex gets endocytosed and becomes internalized into the lysosome to undergo degeneration.^{34,35} Presence of PCSK9 enhances degradation of LDL-R, therefore leading to fewer number of LDL-R present on the cell surface.^{34,35} Proprotein convertase subtilisin/kexin type 9 circulating in the plasma can bind to LDL-R in various organ systems such as the liver, intestines, kidneys, lungs, pancreas, and adipose tissue.^{36,37} This mechanism explains an inverse relationship between PCSK9 concentrations in plasma and LDL-R.^{38–40} Increase in the plasma PCSK9 concentration results in an elevated LDL-C concentration because uptake of LDL-C becomes reduced due to a decreased number of LDL-R on the cell surface (Figure 1A). There are more complicated interactions between PCSK9, LDL-R, LDL-C, and statins, which is beyond the perspective of this review and has recently been published elsewhere.⁴¹

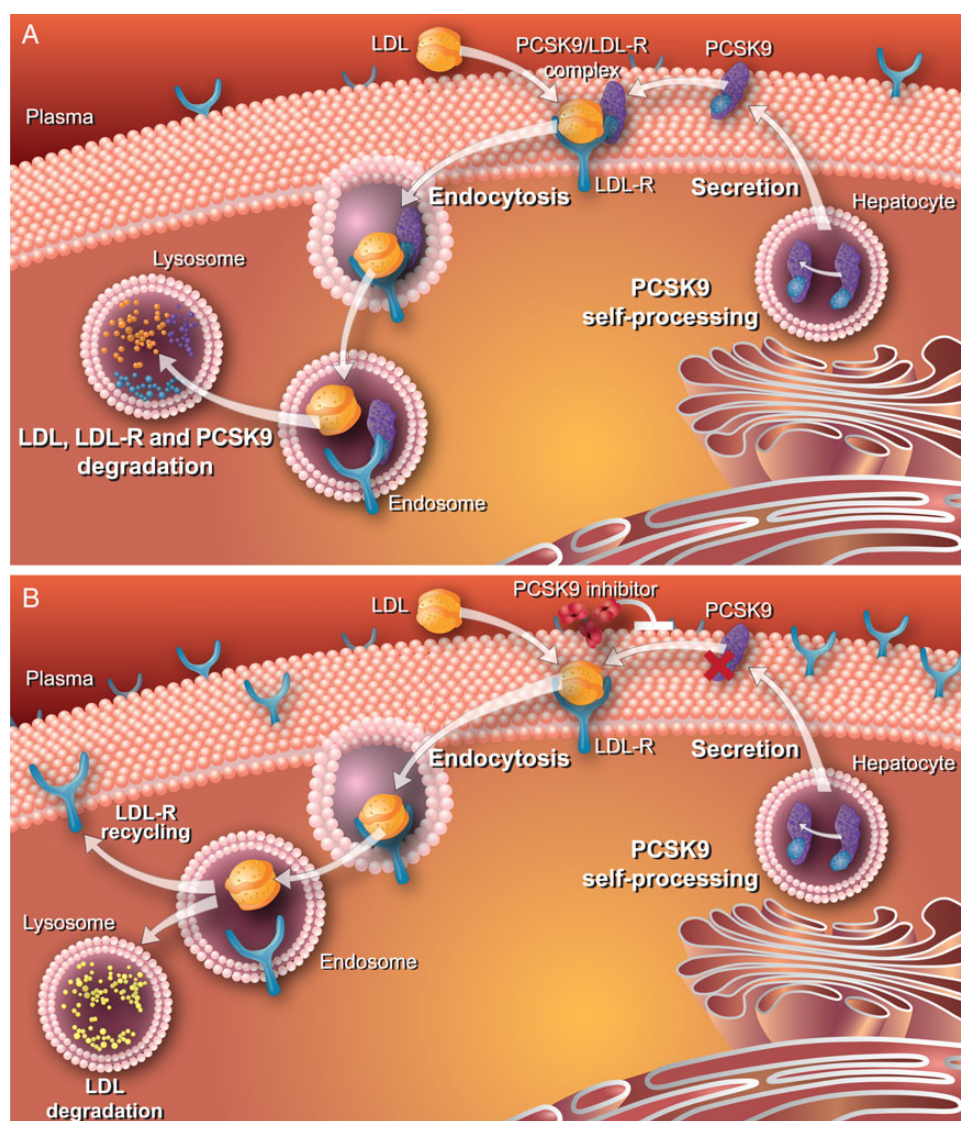


Figure 1 (A) Regulation of hepatic LDL receptor expression by proprotein convertase subtilisin/kexin type 9. (B) Mechanism of low-density lipoprotein cholesterol reduction by proprotein convertase subtilisin/kexin type 9 inhibition. (Adopted from Dadu et al.⁷⁰).

Effects of proprotein convertase subtilisin/kexin type 9 inhibition on low-density lipoprotein cholesterol concentration

When the activity of PCSK9 is inhibited, a larger number of LDL-R returns to the cell surface instead of being degraded in the lysosome. A consequence is increased uptake of LDL-C from the bloodstream. PCSK9 inhibitors decrease LDL-C concentrations through this mechanism (Figure 1B).

Clinical studies on proprotein convertase subtilisin/kexin type 9 inhibition

Pharmacokinetics and pharmacodynamics of proprotein convertase subtilisin/kexin type 9 inhibitors

The pharmacokinetics of PCSK9 inhibitor is represented in Figure 2. Prior to administration of a PCSK9 inhibitor, PCSK9 molecules are freely circulating in plasma (or already bound to the LDL-C receptor). Immediately after an injection of anti-PCSK9 antibody, the antibody binds free PCSK9, and plasma free PCSK9 gets rapidly depleted. Because free PCSK9 concentrations are reduced, fewer LDL-R become degraded in the lysosomal compartments. This results in larger number of LDL-R on the cell surface, and in turn, more LDL particles become bound to LDL-R. Low-density lipoprotein cholesterol concentration gets subsequently reduced. As PCSK9 antibody gets metabolized and its concentration lowered with time, increase in concentration of plasma free PCSK9 occurs followed by recovery of LDL-C concentrations, because more LDL-R become bound to PCSK9 leading to increased degeneration of LDL-R in the lysosomal compartment. This effect of PCSK9 inhibition on LDL-C has been confirmed in a Phase 1 study with healthy volunteers.⁴²

Completed clinical trials

Effects on low-density lipoprotein cholesterol

Table 2 summarizes results from the published Phase 3 trials. Proprotein convertase subtilisin/kexin type 9 inhibitors significantly reduced LDL-C compared with the active comparator or placebo in each PCSK9 inhibitor, patient population, background treatment, and treatment arm. This effect on LDL-C appears to be dose dependent as shown in Figure 3.^{43,44} The effects of PCSK9 inhibition in LDL-C are not affected by age, gender, region, body mass index, and LDL-C concentrations.

Effects of background therapy with statins

As PCSK9 inhibitors would be used concomitantly with statins in most cases, a question arises as to whether PCSK9 inhibitors have different effects depending on background statin use. When evolocumab was administered in 56 healthy volunteers and 57 individuals on statin, there was no difference in the amplitude of LDL-C reduction.⁴⁵ Similarly, in a Phase 2 study with alirocumab, reduction in LDL-C concentrations were not affected by the dose of atorvastatin (10–40 mg daily).⁴³ Another alirocumab study showed no difference in effects between the arm treated with atorvastatin 10 and 80 mg.⁴⁶ The DESCARTES trial provides further insights; the mean reduction in LDL-C in the evolocumab group was similar between patients taking no statin, atorvastatin 10 mg and atorvastatin 80 mg.⁴⁷ Likewise, there was no significant difference in the LDL-lowering effects of the LAPLACE-2 study between different background statin intensity and doses (Figure 4).⁴⁸

Frequency of administration

There have been several Phase 2 trials that assessed LDL-C lowering by PCSK9 inhibitors in different administration frequencies; the effects were similar between 2- and 4-week injection intervals (Figure 4). As shown in Figure 5, however, administration of alirocumab

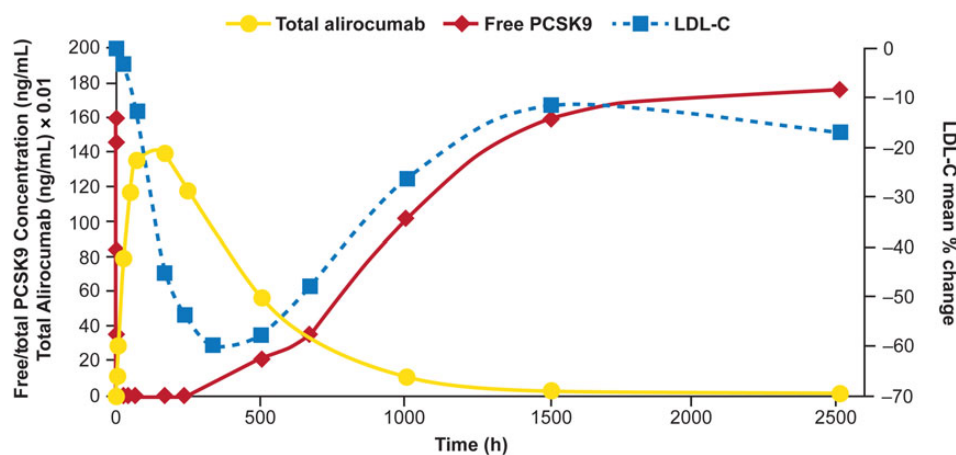


Figure 2 Relationship between anti-proprotein convertase subtilisin/kexin type 9 monoclonal antibody, plasma free proprotein convertase subtilisin/kexin type 9, and low-density lipoprotein cholesterol concentrations (modified from Koren et al.⁷¹).

Table 2 Published Phase 3 clinical trials of monoclonal antibodies against proprotein convertase subtilisin/kexin type 9

Trial name	PCSK9 inhibitor	Population and study design	Number of patients	Study duration	LDL-C	ApoB	Non-HDL-C	TG	HDL-C	Lp(a)
ODYSSEY Mono	Alirocumab	Patients with hypercholesterolaemia on no statins, compared with ezetimibe	103	24 weeks	−31.6	−25.8	−25.5	−1.2	4.4	−4.4
ODYSSEY COMBO I	Alirocumab	Hypercholesterolaemia not adequately controlled (with maximum dose of a statin with or without other lipid-modifying therapy), and high CVD risk	311	24 weeks	−45.9	−35.8	−37.5	−0.6	7.3	−14.6
ODYSSEY COMBO II	Alirocumab	Hypercholesterolaemia not adequately controlled (maximum dose of a statin with or without other lipid-modifying therapy), and high CVD risk	707	24 weeks	−29.7	−22.4	−22.9	−0.3	8.1	−21.7
ODYSSEY LONG TERM	Alirocumab	Hypercholesterolaemia not adequately controlled with current lipid-modifying therapy, and high CVD risk	2341	24 weeks	−61.9	−54.0	−52.3	−17.3	4.6	−25.6
DESCARTES	Evolocumab	Patients with hyperlipidaemia 420 mg Q4W added to diet alone or to diet plus atorvastatin or to diet plus atorvastatin plus ezetimibe	901	52 weeks	−57.0	−44.2	−50.3	−11.5	5.4	−22.4
LAPLACE-2	Evolocumab	Patients with hypercholesterolaemia, 140 mg Q2W or 420 mg Q4W added to moderate- or high-intensity statin therapy, compared with ezetimibe or placebo	2067	12 weeks	−59.2, −70.6	−47.0, −61.4	−54.9, −66.6	−9.3, −31.4	3.2, 9.8	−19.8, −36.5
GAUSS-2	Evolocumab	Patients with statin intolerance, 140 mg Q2W or 420 mg Q4W compared with ezetimibe	307	12 weeks	−68.8, −69.7	−32.9, −33.1			3.6, 4.8	−25.3, −27.9
MENDEL-2	Evolocumab	Patients with hypercholesterolaemia on no statins, 140 mg Q2W or 420 mg Q4W compared with ezetimibe	614	12 weeks	−54.8, −57.1	−47.8, −48.4	−49.8, −51.2	−6.2, −17.7	5.9, 9.3	−17.8, −20.4
RUTHERFORD-2	Evolocumab	Patients with heFH, 140 mg Q2W or 420 mg Q4W	329	12 weeks	−59.2, −61.3	−49.1, −49.4	−54.8, 55.0	−11.6, −19.6	9.1, 9.2	−28.2, −31.6
OSLER-2	Evolocumab	Hypercholesterolaemia or mixed dyslipidaemia; completion of previous evolocumab study (no specification regarding statin therapy)	4465	12 weeks	−61	−47.3	−52.0	−12.6	7.0	−25.5
TESLA Part B	Evolocumab	Patients with hoFH, not on apheresis, 420 mg Q4W	49	12 weeks	−30.9	−23.1		0.3	−0.1	−11.8

ApoB, apolipoprotein B; HDL-C, high-density lipoprotein cholesterol; heFH, heterozygous familial hypercholesterolaemia; hoFH, homozygous familial hypercholesterolaemia; Lp(a), lipoprotein(a); Q2W, every 2 weeks; Q4W, every 4 weeks; TG, triglycerides. Results are expressed as percent treatment difference from baseline adjusted for placebo or active comparator. Results are expressed as a range when multiple comparisons were reported within one trial. Modified from Gouni-Berthold *et al.*⁷²

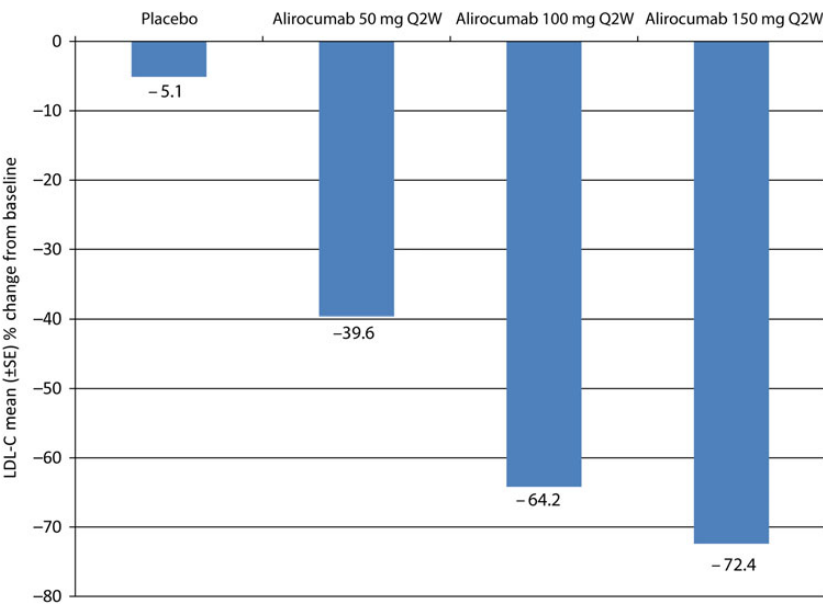


Figure 3 Dose–response relationship between proprotein convertase subtilisin/kexin type 9 inhibitors and mean percent change in low-density lipoprotein cholesterol concentrations with alirocumab (modified from McKenney *et al.*⁴³).

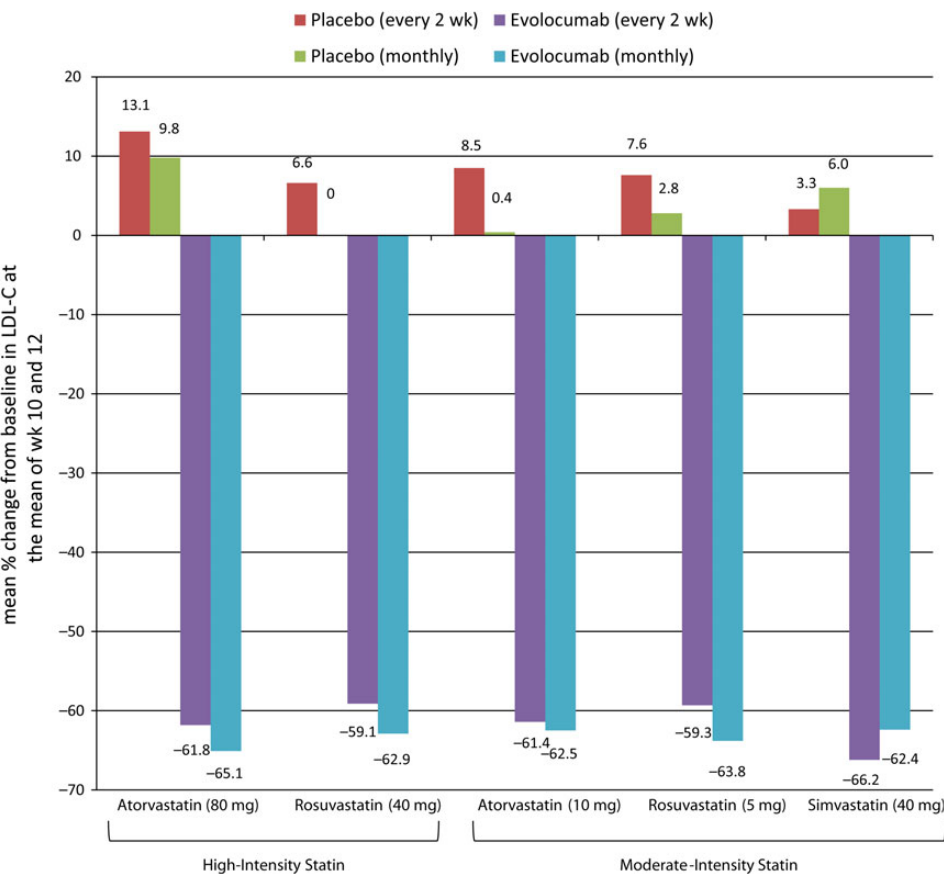


Figure 4 Low-density lipoprotein-lowering effects of proprotein convertase subtilisin/kexin type 9 inhibitor by different statin potency and doses (modified from Robinson *et al.*⁴⁸).

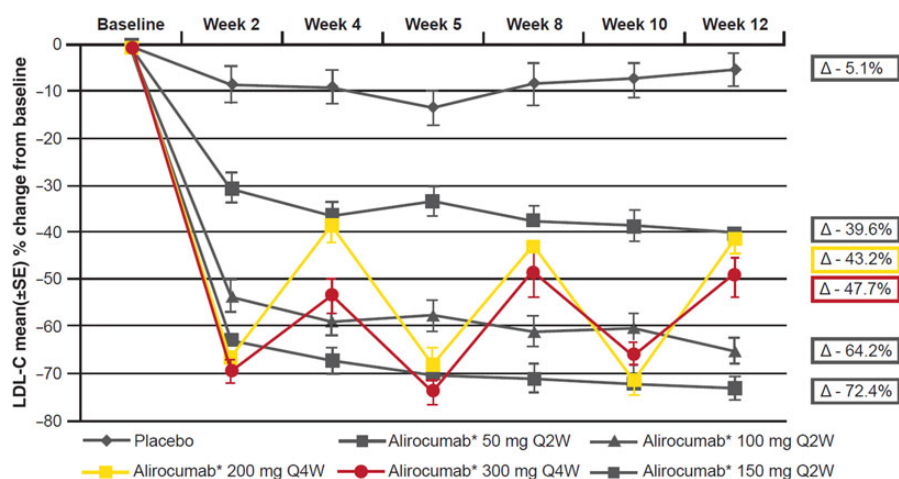


Figure 5 Change in low-density lipoprotein cholesterol with different alirocumab concentrations and intervals from baseline to week 12 (figure modified from McKenney et al.⁴³).

every 4 weeks resulted in a larger fluctuation of LDL-C concentrations than every 2 weeks.⁴³

Safety of proprotein convertase subtilisin/kexin type 9 administration

In terms of the safety with PCSK9 inhibitors, 36–75% of the patients on PCSK9 inhibitors in Phase 3 trials had adverse events and 2–6% had serious adverse events.^{47–49} Adverse events leading to the discontinuation of the study drug occurred in 2–10%. These numbers were similar to the comparator in each study. Elevation of liver aminotransferase concentrations > 3 times of the upper normal limit in patients receiving PCSK9 inhibitor was observed in <2% of the patients. Abnormal elevation of creatinine kinase concentrations was similarly rare. No neutralizing anti-drug antibody, which may impact pharmacokinetics of PCSK9 inhibitors, effects on LDL-C concentrations or safety, was reported in any of the Phase 3 trials.

Tolerability

In the ODYSSEY MONO trials, injection site reaction was observed in 1 of 52 patients who received alirocumab compared with 2 of 51 in placebo at 24 weeks.⁴⁹ In the OSLER trial with a 52-week follow-up, only 1 of 38 patients who experienced injection-site reaction discontinued evolocumab.⁵⁰ Similarly, the DESCARTES study reported that 34 of 599 patients (5.7%) in the evolocumab group and 15 of 302 (5.0%) in the placebo group had injection-site reactions, and only 1 patient discontinued evolocumab.⁴⁷ In the ODYSSEY COMBO II study, ~85% of patients were still taking study medication (or placebo) at 1 year, indicating good acceptance and tolerability of parenteral LDL-C-lowering therapies.⁵¹

Target patient populations

Familial hypercholesterolaemia

Heterozygous FH is one of the most common genetic diseases (prevalence 1/200–1/500) and characterized by extremely high

LDL-C concentrations. This results in development of atherosclerosis and cardiovascular disease in younger age.⁵² The mean age of onset of cardiovascular disease was 42–46 years in men and 51–52 years in women.^{53,54} Homozygous FH is a more severe form of FH, and these patients develop cardiovascular disease at mean age of 20 years.⁵⁵ The treatment goal for adult patients with heterozygous FH who also have coronary heart disease or diabetes is 70 mg/dL.⁵² However, ~80% of adult patients with heterozygous FH on lipid-lowering treatment do not reach the LDL-C goal of 100 mg/dL.⁵⁶ Therefore, FH patients appear to be a good target population for PCSK9 inhibitors.⁵⁷

Patients who cannot reach low-density lipoprotein cholesterol goals with high-dose statins

Patients with severe non-FH and FH sometimes cannot achieve adequately low LDL-C concentrations despite maximum dose of highly potent statins. As a number of studies have shown that the incidence of cardiovascular events directly correlates with LDL-C concentrations,⁵⁸ those with high LDL-C despite maximum statin therapy is at increased risk of developing cardiovascular events. The guidelines do not make recommendations as to which additional lipid-lowering agents should be chosen for these patients. Proprotein convertase subtilisin/kexin type 9 inhibitors are known to reduce LDL-C concentrations even when added to background statin therapy, and may offer additional protective effects against cardiovascular complications. This would probably have a substantial clinical impact as the IMPROVE-IT study demonstrated achieving lower LDL-C targets than the conventional goal, from 70 to 54 mg/dL, resulted in a significant reduction in adverse cardiovascular events.⁵⁹

Statin intolerance

Statins are most often the first line treatment for hypercholesterolaemia, and have been shown to reduce LDL-C concentrations and cardiovascular complications. However, some patients cannot tolerate statin therapy because of various reasons such as myalgia,

Table 3 Selected Phase 3 trials on monoclonal antibody against proprotein convertase subtilisin/kexin type 9 that are (A) ongoing and (B) completed and presented but not published as a full manuscript

Name of study	Monoclonal antibody	Patient population	Study objectives and follow-up period	Target completion date
(A)				
GLAGOV	Evolocumab	Coronary heart disease; clinical indication for coronary catheterization; and LDL-C level ≥ 80 mg/dL or, with additional risk factors, ≥ 60 and < 80 mg/dL (no specification regarding statin therapy)	To determine the effects of evolocumab every 4 weeks on atherosclerotic disease burden (per cent atheroma volume measured by intravascular ultrasonography), at 72 weeks	November 16
FOURIER	Evolocumab	Clinical CVD, high risk of recurrent CVD event, and LDL-C level ≥ 70 mg/dL or non-HDL-C ≥ 100 mg/dL (no specification regarding statin therapy)	To assess the effect of evolocumab every 2 or 4 weeks plus a statin vs. placebo plus a statin on major CVD events (CVD death, nonfatal myocardial infarction, unstable angina requiring hospitalization, stroke, or coronary revascularization), at 5 years	February 18
TAUSSIG	Evolocumab	Homozygous FH or PCSK9 mutations; LDL-C level above ATP III target or receiving apheresis; and completion of previous evolocumab study (no specification regarding statin therapy)	To assess the long-term safety and efficacy of evolocumab every 2 or 4 weeks on LDL-C level in patients with severe FH, at 5 years	January 20
SPIRE-HF	Bococizumab	Heterozygous FH; high or very high CVD risk; LDL-C level > 70 mg/dL and Tg level ≤ 400 mg/dL (with statin therapy)	To compare the effect of bococizumab and a statin vs. placebo and a statin on LDL-C level in patients with heterozygous FH, at 12 weeks	May 16
SPIRE-HR	Bococizumab	High or very high CVD risk; LDL-C level > 70 mg/dL and Tg level ≤ 400 mg/dL (with statin therapy)	To compare the effect of bococizumab and a statin vs. placebo and a statin on LDL-C level, at 12 weeks	April 16
SPIRE-LDL	Bococizumab	High or very high CVD risk; LDL-C level > 70 mg/dL and Tg level ≤ 400 mg/dL (with statin therapy)	To compare the effect of bococizumab and a statin vs. placebo and a statin on LDL-C level, at 12 weeks	June 16
SPIRE-1	Bococizumab	High CVD risk; LDL-C level ≥ 70 mg/dL and < 100 mg/dL, or non-HDL-C level ≥ 100 mg/dL and < 130 mg/dL, with lipid-lowering therapy (no specification regarding statin therapy)	To compare the effect of bococizumab vs. placebo on reducing the occurrence of major cardiovascular events, including cardiovascular death, myocardial infarction, stroke, and unstable angina requiring urgent revascularization, at 5 years	August 17
SPIRE-2	Bococizumab	High CVD risk; LDL-C level ≥ 100 mg/dL or non-HDL-C level ≥ 130 mg/dL, with lipid-lowering therapy (no specification regarding statin therapy)	To compare the effect of bococizumab vs. placebo on reducing the occurrence of major cardiovascular events, including cardiovascular death, myocardial infarction, stroke, and unstable angina requiring urgent revascularization, at 5 years	August 17
ODYSSEY OUTCOMES	Alirocumab	Recent (in the past 4–16 weeks) acute coronary syndrome event requiring hospitalization	To compare the effect of alirocumab vs. placebo on CVD events (cardiovascular death, nonfatal myocardial infarction, fatal and non-fatal ischaemic stroke, and unstable angina requiring hospitalization), for up to 64 months	January 18
(B)				
ODYSSEY ALTERNATIVE	Alirocumab	Statin intolerance; primary hypercholesterolaemia (heterozygous FH or non-FH); and moderate, high, or very high CVD risk (no statin therapy)	To evaluate the efficacy and safety of alirocumab vs. ezetimibe and vs. atorvastatin, after 24 weeks of treatment	Completed; preliminary results presented at AHA Scientific Sessions 2014

Continued

Table 3 Continued

Name of study	Monoclonal antibody	Patient population	Study objectives and follow-up period	Target completion date
ODYSSEY OPTIONS I	Alirocumab	Hypercholesterolaemia (heterozygous FH or non-FH) not adequately controlled (atorvastatin with or without other lipid-modifying therapy), and high CVD risk	To evaluate the reduction in LDL-C level by alirocumab as an add-on therapy to atorvastatin, vs. ezetimibe as an add-on therapy to atorvastatin, vs. doubling the atorvastatin dose, and vs. switching from atorvastatin to rosuvastatin, after 24 weeks of treatment	Completed; preliminary results presented at AHA Scientific Sessions 2014
ODYSSEY OPTIONS II	Alirocumab	Hypercholesterolaemia not adequately controlled (rosuvastatin with or without other lipid-modifying therapy), and high CVD risk	To evaluate the reduction in LDL-C level by alirocumab as an add-on therapy to rosuvastatin, vs. ezetimibe as an add-on therapy to rosuvastatin, and vs. doubling the rosuvastatin dose, after 24 weeks of treatment	Completed; preliminary results presented at AHA Scientific Sessions 2014
ODYSSEY FH I	Alirocumab	Heterozygous FH not adequately controlled with current lipid-modifying therapy (no specification regarding statin therapy)	To evaluate the effect of alirocumab vs. placebo on LDL-C level after 24 weeks of treatment (including background statin or other lipid-modifying therapy)	Completed; preliminary results presented at ESC Congress 2014
ODYSSEY FH II	Alirocumab	Heterozygous FH not adequately controlled (with maximally tolerated statin with or without other lipid-modifying therapy)	To demonstrate the reduction in LDL-C level by alirocumab vs. placebo as an add-on therapy to stable, maximally tolerated daily statin (atorvastatin, rosuvastatin, or simvastatin) therapy, with or without other lipid-modifying therapy, after 24 weeks of treatment	Completed; preliminary results presented at AHA Scientific Sessions 2014

elevated liver enzymes, and rhabdomyolysis. The exact prevalence of statin intolerance is unknown and largely depends on how intolerance is defined. However, it has been reported that 12% of patients started on statin had to discontinue the therapy, within which 62% was due to adverse effects.⁶⁰ Patients who cannot tolerate statins remain at high risk of developing cardiovascular events. Nevertheless, monotherapy with other lipid-lowering therapies is not shown to reduce hard clinical endpoints such as coronary artery disease or cardiovascular death. Proprotein convertase subtilisin/kexin type 9 inhibitors are expected to offer an alternative treatment option for these patients with statin intolerance.

Ongoing clinical trials

A number of Phase 3 clinical trials are being conducted for PCSK9 inhibitors such as evolocumab, bococizumab, and alirocumab. Table 3 summarizes the patient population, information on background statin therapy, active comparator vs. placebo, duration of the study and estimated completion time of selected Phase 3 trials that are either ongoing (<https://clinicaltrials.gov/ct2/show/NCT01663402>, <https://clinicaltrials.gov/ct2/show/study/NCT01975376?term=SPIRE-1&rank=1>, <https://clinicaltrials.gov/ct2/show/NCT01968967>, <https://clinicaltrials.gov/ct2/show/NCT01968954>, <https://clinicaltrials.gov/ct2/show/NCT01968980>, <https://clinicaltrials.gov/ct2/show/NCT01624142>, <https://clinicaltrials.gov/ct2/show/NCT01764633>, <https://clinicaltrials.gov/ct2/show/NCT01813422>) or completed but unpublished.^{61–63} Similar to what has been performed in the past in relation to statin studies, it would be of clinical and scientific interest to assess

the effects of PCSK9 inhibition onto the progression or regression of coronary atherosclerosis by utilizing coronary imaging techniques. In this context, the GLAGOV trial is currently ongoing to assess change in per cent atheroma volume as determined by intravascular ultrasound at 78 weeks post-randomization (<https://clinicaltrials.gov/ct2/show/NCT01813422>).

Challenges

Cost effectiveness

The cost of a cardiovascular event is high; the total first year cost after an acute coronary event is ~\$22 529 (~19 800 euros).⁶⁴ Among patients without FH, high-dose statins have shown to be cost effective for reducing cardiovascular disease in long term.⁶⁵ No prior study has performed cost-effectiveness analysis in terms of PCSK9 inhibitor compared with previously existing treatment for hyperlipidaemia. There appears to be a specific target for PCSK9 inhibitors to be cost-effective; among patients with FH who require LDL apheresis, the treatment is limited by the high cost that can reach between \$45 000 and \$100 000 (~39 600–88 000 euros) per year.⁶⁶ Especially, in this high-risk population with large health-care costs, PCSK9 inhibitors may prove to be cost effective. The cost-effectiveness of PCSK9 inhibitors would also depend on the marketing price and the patient's baseline risk of cardiovascular events.⁶⁷

Compliance with injections

Compliance issues associated with injectable medications do not often get addressed by randomized controlled trials, but could pose a significant limitation in daily clinical practice when utilizing

PCSK9 inhibitors. There are mainly two characteristics of PCSK9 inhibitors that can significantly affect patient compliance. First, practically speaking, subcutaneous self-injections would be challenging for some of the patients; however, if the patient cannot perform injection at home, the patient would have to visit the clinic at least monthly. Second, the amount of medication to be injected is rather large; for example, 6 mL of evolocumab needed to be injected monthly in the clinical trials.⁴⁷ Although this could be divided into 3 mL injection twice or 2 mL injection three times, pain associated with subcutaneous injection of this amount of liquid can be substantial. The positive news is that several Phase 3 trials reported that most patients are tolerating subcutaneous injection of the study drug (or placebo) at 52 weeks.⁵¹ A post-marketing registry data would help understand what proportion of patients can actually remain compliant with monthly self-injections in the real world.

Long-term safety

Safety of PCSK9 inhibitors in longer term than 12 weeks has been reported only in three published clinical trials. The first trial, ODYSSEY MONO, was a medium-sized trial (~50 patients in each arm) and reported no significant increase in adverse events in the PCSK9 inhibitor arm. The second trial, DESCARTES study, was a larger trial with longer follow-up period of 52 weeks involving 900 patients. The rates of adverse events were similar in the evolocumab arm and the placebo arm.⁴⁷ The third was the OSLER trial.⁵⁰ At 52 weeks, adverse events were observed in 81% of patients who received evolocumab compared with 73% in the standard of care group. In the evolocumab arm, 3.7% had adverse events leading to discontinuation of the drug. Regarding safety and tolerability in >1 year, alirocumab was tolerated well without a significant increase in adverse effects attributable to the study drug according to the data from the ODYSSEY LONG TERM trial.⁶⁸ A number of Phase 3 trials with planned follow-up period >1 year are ongoing, such as the ODYSSEY OUTCOMES (<https://clinicaltrials.gov/ct2/show/NCT01663402>), GLAGOV (<https://clinicaltrials.gov/ct2/show/NCT01813422>), FOURIER (<https://clinicaltrials.gov/ct2/show/NCT01764633>), TAUSIG (<https://clinicaltrials.gov/ct2/show/NCT01624142>), and SPIRE 1/2 trials (<https://clinicaltrials.gov/ct2/show/NCT01975389?term=SPIRE-2&rank=1>), (<https://clinicaltrials.gov/ct2/show/study/NCT01975376?term=SPIRE-1&rank=1>).

Conclusions

In summary, PCSK9 has experienced a rapid evolution from genetic discovery, to drug target, to animal, then human testing, and now to large clinical outcomes trials. It has provided new biologic insights and a potent natural pharmacologic mechanism to further lower LDL-C. If outcomes of the ongoing trials are positive, they would shift the guidelines for treatment of hypercholesterolaemia, and may result in adding back of the LDL-based treatment decisions in the guidelines. With 50–70% further reduction in LDL-C, PCSK9 inhibitors appear to be a promising pharmacologic intervention for patients with difficult-to-treat hypercholesterolaemia.

Conflict of interest: Relationship with industry: Y.J.S. has no conflict of interest related to this article. C.P.C. reports grants from Accumetrics, Arisaph, Astra Zeneca, Boehringer-Ingelheim, and

Janssen; grants and consulting fees from GlaxoSmithKline, Merck, and Takeda; consulting fees from BMS, CSL Behring, Essentialis, Lipimedix, Pfizer, Regeneron, Sanofi, and Kowa.

References

- Agarwal SK, Avery CL, Ballantyne CM, Catellier D, Nambi V, Saunders J, Sharrett AR, Coresh J, Heiss G, Hoogeveen RC. Sources of variability in measurements of cardiac troponin T in a community-based sample: the atherosclerosis risk in communities study. *Clin Chem* 2011;**57**:891–897.
- Boekholdt SM, Arsenault BJ, Mora S, Pedersen TR, LaRosa JC, Nestel PJ, Simes RJ, Durrington P, Hitman GA, Welch KM, DeMicco DA, Zwinderman AH, Clearfield MB, Downs JR, Tonkin AM, Colhoun HM, Gotto AM Jr, Ridker PM, Kastelein JJ. Association of LDL cholesterol, non-HDL cholesterol, and apolipoprotein B levels with risk of cardiovascular events among patients treated with statins: a meta-analysis. *JAMA* 2012;**307**:1302–1309.
- Sniderman AD, Williams K, Contois JH, Monroe HM, McQueen MJ, de Graaf J, Furberg CD. A meta-analysis of low-density lipoprotein cholesterol, non-high-density lipoprotein cholesterol, and apolipoprotein B as markers of cardiovascular risk. *Circ Cardiovasc Qual Outcomes* 2011;**4**:337–345.
- Baigent C, Keech A, Kearney PM, Blackwell L, Buck G, Pollicino C, Kirby A, Sourjina T, Peto R, Collins R, Simes R. Cholesterol Treatment Trialists' (CTT) Collaborators. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;**366**:1267–1278.
- Genser B, Marz W. Low density lipoprotein cholesterol, statins and cardiovascular events: a meta-analysis. *Clin Res Cardiol* 2006;**95**:393–404.
- Avis HJ, Hutten BA, Gagne C, Langslet G, McCrindle BW, Wiegman A, Hsia J, Kastelein JJ, Stein EA. Efficacy and safety of rosuvastatin therapy for children with familial hypercholesterolemia. *J Am Coll Cardiol* 2010;**55**:1121–1126.
- Seidah NG, Prat A. Precursor convertases in the secretory pathway, cytosol and extracellular milieu. *Essays Biochem* 2002;**38**:79–94.
- Seidah NG, Benjannet S, Wickham L, Marcinkiewicz J, Jasmin SB, Stifani S, Basak A, Prat A, Chretien M. The secretory proprotein convertase neural apoptosis-regulated convertase 1 (NARC-1): liver regeneration and neuronal differentiation. *Proc Natl Acad Sci USA* 2003;**100**:928–933.
- Naureckiene S, Ma L, Sreekumar K, Purandare U, Lo CF, Huang Y, Chiang LW, Grenier JM, Ozenberger BA, Jacobsen JS, Kennedy JD, DiStefano PS, Wood A, Bingham B. Functional characterization of NARC 1, a novel proteinase related to proteinase K. *Arch Biochem Biophys* 2003;**420**:55–67.
- Benjannet S, Rhainds D, Essalmani R, Mayne J, Wickham L, Jin W, Asselin MC, Hamelin J, Varret M, Allard D, Trillard M, Abifadel M, Tebon A, Attie AD, Rader DJ, Boileau C, Brissette L, Chretien M, Prat A, Seidah NG. NARC-1/PCSK9 and its natural mutants: zymogen cleavage and effects on the low density lipoprotein (LDL) receptor and LDL cholesterol. *J Biol Chem* 2004;**279**:48865–48875.
- Cunningham D, Danley DE, Geoghegan KF, Griffor MC, Hawkins JL, Subashi TA, Varghese AH, Ammirati MJ, Culp JS, Hoth LR, Mansour MN, McGrath KM, Seddon AP, Shenolikar S, Stutzman-Engwall KJ, Warren LC, Xia D, Qiu X. Structural and biophysical studies of PCSK9 and its mutants linked to familial hypercholesterolemia. *Nat Struct Mol Biol* 2007;**14**:413–419.
- Abifadel M, Varret M, Rabes JP, Allard D, Ouguerram K, Devillers M, Cruaud C, Benjannet S, Wickham L, Erlich D, Derre A, Villegier L, Farnier M, Beucier I, Bruckert E, Chambaz J, Chanu B, Lecerf JM, Luc G, Moulin P, Weissenbach J, Prat A, Krempf M, Junien C, Seidah NG, Boileau C. Mutations in PCSK9 cause autosomal dominant hypercholesterolemia. *Nat Genet* 2003;**34**:154–156.
- Allard D, Amsellem S, Abifadel M, Trillard M, Devillers M, Luc G, Krempf M, Reznik Y, Girardet JP, Fredenrich A, Junien C, Varret M, Boileau C, Benlian P, Rabes JP. Novel mutations of the PCSK9 gene cause variable phenotype of autosomal dominant hypercholesterolemia. *Hum Mutat* 2005;**26**:497.
- Humphries SE, Cranston T, Allen M, Middleton-Price H, Fernandez MC, Senior V, Hawe E, Iversen A, Wray R, Crook MA, Wierzbicki AS. Mutational analysis in UK patients with a clinical diagnosis of familial hypercholesterolaemia: relationship with plasma lipid traits, heart disease risk and utility in relative tracing. *J Mol Med (Berl)* 2006;**84**:203–214.
- Maxwell KN, Breslow JL. Proprotein convertase subtilisin kexin 9: the third locus implicated in autosomal dominant hypercholesterolemia. *Curr Opin Lipidol* 2005;**16**:167–172.
- Humphries SE, Whittall RA, Hubbart CS, Maplebeck S, Cooper JA, Soutar AK, Naoumova R, Thompson GR, Seed M, Durrington PN, Miller JP, Betteridge DJ, Neil HA. Simon Broome Familial Hyperlipidaemia Register Group and Scientific Steering Committee. Genetic causes of familial hypercholesterolaemia in patients in the UK: relation to plasma lipid levels and coronary heart disease risk. *J Med Genet* 2006;**43**:943–949.

17. Cohen J, Pertsemlidis A, Kotowski IK, Graham R, Garcia CK, Hobbs HH. Low LDL cholesterol in individuals of African descent resulting from frequent nonsense mutations in PCSK9. *Nat Genet* 2005;**37**:161–165.
18. Cohen JC, Boerwinkle E, Mosley TH Jr, Hobbs HH. Sequence variations in PCSK9, low LDL, and protection against coronary heart disease. *N Engl J Med* 2006;**354**:1264–1272.
19. Benn M, Nordestgaard BG, Grande P, Schnohr P, Tybjaerg-Hansen A. PCSK9 R46L, low-density lipoprotein cholesterol levels, and risk of ischemic heart disease: 3 independent studies and meta-analyses. *J Am Coll Cardiol* 2010;**55**:2833–2842.
20. Hooper AJ, Marais AD, Tanyanyiwa DM, Burnett JR. The C679X mutation in PCSK9 is present and lowers blood cholesterol in a Southern African population. *Atherosclerosis* 2007;**193**:445–448.
21. Huang CC, Fornage M, Lloyd-Jones DM, Wei GS, Boerwinkle E, Liu K. Longitudinal association of PCSK9 sequence variations with low-density lipoprotein cholesterol levels: the Coronary Artery Risk Development in Young Adults Study. *Circ Cardiovasc Genet* 2009;**2**:354–361.
22. Folsom AR, Peacock JM, Boerwinkle E, Atherosclerosis Risk in Communities (ARIC) Study Investigators. Variation in PCSK9, low LDL cholesterol, and risk of peripheral arterial disease. *Atherosclerosis* 2009;**202**:211–215.
23. Cariou B, Ougueram K, Zair Y, Guerois R, Langhi C, Kourimate S, Benoit I, Le May C, Gayet C, Belabbas K, Dufernez F, Chetiveaux M, Tarugi P, Krempf M, Benlian P, Costet P. PCSK9 dominant negative mutant results in increased LDL catabolic rate and familial hypobetalipoproteinemia. *Arterioscler Thromb Vasc Biol* 2009;**29**:2191–2197.
24. Zhao Z, Tuakli-Wosornu Y, Lagace TA, Kinch L, Grishin NV, Horton JD, Cohen JC, Hobbs HH. Molecular characterization of loss-of-function mutations in PCSK9 and identification of a compound heterozygote. *Am J Hum Genet* 2006;**79**:514–523.
25. Chan JC, Piper DE, Cao Q, Liu D, King C, Wang W, Tang J, Liu Q, Higbee J, Xia Z, Di Y, Shetterly S, Arimura Z, Salomonis H, Romanow WG, Thibault ST, Zhang R, Cao P, Yang XP, Yu T, Lu M, Retter MW, Kwon G, Henne K, Pan O, Tsai MM, Fuchslocher B, Yang E, Zhou L, Lee KJ, Daris M, Sheng J, Wang Y, Shen WD, Yeh WC, Emery M, Walker NP, Shan B, Schwarz M, Jackson SM. A proprotein convertase subtilisin/kexin type 9 neutralizing antibody reduces serum cholesterol in mice and nonhuman primates. *Proc Natl Acad Sci USA* 2009;**106**:9820–9825.
26. Zhang L, McCabe T, Condra JH, Ni YG, Peterson LB, Wang W, Strack AM, Wang F, Pandit S, Hammond H, Wood D, Lewis D, Rosa R, Mendoza V, Cumiskey AM, Johns DG, Hansen BC, Shen X, Geoghagen N, Jensen K, Zhu L, Wietsche K, Wisniewski D, Huang L, Zhao JZ, Ernst R, Hampton R, Haytko P, Ansbro F, Chlewicki S, Chin J, Mitnaul LJ, Pellacani A, Sparrow CP, An Z, Strohl W, Hubbard B, Plump AS, Blom D, Sitlani A. An anti-PCSK9 antibody reduces LDL-cholesterol on top of a statin and suppresses hepatocyte SREBP-regulated genes. *Int J Biol Sci* 2012;**8**:310–327.
27. Ni YG, Di Marco S, Condra JH, Peterson LB, Wang W, Wang F, Pandit S, Hammond HA, Rosa R, Cummings RT, Wood DD, Liu X, Bottomley MJ, Shen X, Cubbon RM, Wang SP, Johns DG, Volpari C, Hamuro L, Chin J, Huang L, Zhao JZ, Vitelli S, Haytko P, Wisniewski D, Mitnaul LJ, Sparrow CP, Hubbard B, Carfi A, Sitlani A. A PCSK9-binding antibody that structurally mimics the EGF(A) domain of LDL-receptor reduces LDL cholesterol in vivo. *J Lipid Res* 2011;**52**:78–86.
28. Frank-Kamenetsky M, Grefhorst A, Anderson NN, Racie TS, Bramlage B, Akinc A, Butler D, Charisse K, Dorkin R, Fan Y, Gamba-Vitalo C, Hadwiger P, Jayaraman M, John M, Jayaprakash KN, Maier M, Nechev L, Rajeev KG, Read T, Rohl I, Soutschek J, Tan P, Wong J, Wang G, Zimmermann T, de Fougerolles A, Vornlocher HP, Langer R, Anderson DG, Manoharan M, Kotliansky V, Horton JD, Fitzgerald K. Therapeutic RNAi targeting PCSK9 acutely lowers plasma cholesterol in rodents and LDL cholesterol in nonhuman primates. *Proc Natl Acad Sci USA* 2008;**105**:11915–11920.
29. Fitzgerald K, Frank-Kamenetsky M, Shulga-Morskaya S, Liebow A, Bettencourt BR, Sutherland JE, Hutabarat RM, Clausen VA, Karsten V, Cehelsky J, Nochur SV, Kotlianski V, Horton J, Mant T, Chiesa J, Ritter J, Munisamy M, Vaishnav AK, Gollob JA, Simon A. Effect of an RNA interference drug on the synthesis of proprotein convertase subtilisin/kexin type 9 (PCSK9) and the concentration of serum LDL cholesterol in healthy volunteers: a randomised, single-blind, placebo-controlled, phase 1 trial. *Lancet* 2014;**383**:60–68.
30. Gupta N, Fisker N, Asselin MC, Lindholm M, Rosenbohm C, Orum H, Elmen J, Seidah NG, Straarup EM. A locked nucleic acid antisense oligonucleotide (LNA) silences PCSK9 and enhances LDLR expression in vitro and in vivo. *PLoS ONE* 2010;**5**:e10682.
31. Graham MJ, Lemonidis KM, Whipple CP, Subramaniam A, Monia BP, Crooke ST, Crooke RM. Antisense inhibition of proprotein convertase subtilisin/kexin type 9 reduces serum LDL in hyperlipidemic mice. *J Lipid Res* 2007;**48**:763–767.
32. Lindholm MW, Elmen J, Fisker N, Hansen HF, Persson R, Moller MR, Rosenbohm C, Orum H, Straarup EM, Koch T. PCSK9 LNA antisense oligonucleotides induce sustained reduction of LDL cholesterol in nonhuman primates. *Mol Ther* 2012;**20**:376–381.
33. Do RQ, Vogel RA, Schwartz GG. PCSK9 Inhibitors: potential in cardiovascular therapeutics. *Curr Cardiol Rep* 2013;**15**:1–12.
34. Zhang DW, Lagace TA, Garuti R, Zhao Z, McDonald M, Horton JD, Cohen JC, Hobbs HH. Binding of proprotein convertase subtilisin/kexin type 9 to epidermal growth factor-like repeat A of low density lipoprotein receptor decreases receptor recycling and increases degradation. *J Biol Chem* 2007;**282**:18602–18612.
35. Poirier S, Mayer G, Benjannet S, Bergeron E, Marcinkiewicz J, Nassoury N, Mayer H, Nimpf J, Prat A, Seidah NG. The proprotein convertase PCSK9 induces the degradation of low density lipoprotein receptor (LDLR) and its closest family members VLDLR and ApoER2. *J Biol Chem* 2008;**283**:2363–2372.
36. Lagace TA, Curtis DE, Garuti R, McNutt MC, Park SW, Prather HB, Anderson NN, Ho YK, Hammer RE, Horton JD. Secreted PCSK9 decreases the number of LDL receptors in hepatocytes and in livers of parabiotic mice. *J Clin Invest* 2006;**116**:2995–3005.
37. Schmidt RJ, Beyer TP, Bensch WR, Qian YW, Lin A, Kowala M, Alborn WE, Konrad RJ, Cao G. Secreted proprotein convertase subtilisin/kexin type 9 reduces both hepatic and extrahepatic low-density lipoprotein receptors in vivo. *Biochem Biophys Res Commun* 2008;**370**:634–640.
38. Maxwell KN, Fisher EA, Breslow JL. Overexpression of PCSK9 accelerates the degradation of the LDLR in a post-endoplasmic reticulum compartment. *Proc Natl Acad Sci USA* 2005;**102**:2069–2074.
39. Rashid S, Curtis DE, Garuti R, Anderson NN, Bashmakov Y, Ho YK, Hammer RE, Moon YA, Horton JD. Decreased plasma cholesterol and hypersensitivity to statins in mice lacking Pcsk9. *Proc Natl Acad Sci USA* 2005;**102**:5374–5379.
40. Alborn WE, Cao G, Careskey HE, Qian YW, Subramaniam DR, Davies J, Conner EM, Konrad RJ. Serum proprotein convertase subtilisin kexin type 9 is correlated directly with serum LDL cholesterol. *Clin Chem* 2007;**53**:1814–1819.
41. Tavori H, Rashid S, Fazio S. On the function and homeostasis of PCSK9: Reciprocal interaction with LDLR and additional lipid effects. *Atherosclerosis* 2014;**238**:264–270.
42. Stein EA, Mellis S, Yancopoulos GD, Stahl N, Logan D, Smith WB, Lisbon E, Gutierrez M, Webb C, Wu R, Du Y, Kranz T, Gasparino E, Swergold GD. Effect of a monoclonal antibody to PCSK9 on LDL cholesterol. *N Engl J Med* 2012;**366**:1108–1118.
43. McKenney JM, Koren MJ, Kereiakes DJ, Hanotin C, Ferrand AC, Stein EA. Safety and efficacy of a monoclonal antibody to proprotein convertase subtilisin/kexin type 9 serine protease, SAR236553/REGN727, in patients with primary hypercholesterolemia receiving ongoing stable atorvastatin therapy. *J Am Coll Cardiol* 2012;**59**:2344–2353.
44. Giugliano RP, Desai NR, Kohli P, Rogers WJ, Somaratne R, Huang F, Liu T, Mohanavelu S, Hoffman EB, McDonald ST, Abrahamson TE, Wasserman SM, Scott R, Sabatine MS. LAPLACE-TIMI 57 Investigators. Efficacy, safety, and tolerability of a monoclonal antibody to proprotein convertase subtilisin/kexin type 9 in combination with a statin in patients with hypercholesterolaemia (LAPLACE-TIMI 57): a randomised, placebo-controlled, dose-ranging, phase 2 study. *Lancet* 2012;**380**:2007–2017.
45. Dias CS, Shaywitz AJ, Wasserman SM, Smith BP, Gao B, Stolman DS, Crispino CP, Smirnakis KV, Emery MG, Colbert A, Gibbs JP, Retter MW, Cooke BP, Uy ST, Matson M, Stein EA. Effects of AMG 145 on low-density lipoprotein cholesterol levels: results from 2 randomized, double-blind, placebo-controlled, ascending-dose phase 1 studies in healthy volunteers and hypercholesterolemic subjects on statins. *J Am Coll Cardiol* 2012;**60**:1888–1898.
46. Roth EM, McKenney JM, Hanotin C, Asset G, Stein EA. Atorvastatin with or without an antibody to PCSK9 in primary hypercholesterolemia. *N Engl J Med* 2012;**367**:1891–1900.
47. Blom DJ, Hala T, Bolognese M, Lillestol MJ, Toth PD, Burgess L, Ceska R, Roth E, Koren MJ, Ballantyne CM, Monsalvo ML, Tsirtsonis K, Kim JB, Scott R, Wasserman SM, Stein EA. DESCARTES Investigators. A 52-week placebo-controlled trial of evolocumab in hyperlipidemia. *N Engl J Med* 2014;**370**:1809–1819.
48. Robinson JG, Nedergaard BS, Rogers WJ, Fialkow J, Neutel JM, Ramstad D, Somaratne R, Legg JC, Nelson P, Scott R, Wasserman SM, Weiss R. LAPLACE-2 Investigators. Effect of evolocumab or ezetimibe added to moderate- or high-intensity statin therapy on LDL-C lowering in patients with hypercholesterolemia: the LAPLACE-2 randomized clinical trial. *JAMA* 2014;**311**:1870–1882.
49. Roth EM, Taskinen MR, Ginsberg HN, Kastelein JJ, Colhoun HM, Robinson JG, Merlet L, Pordy R, Baccara-Dinet MT. Monotherapy with the PCSK9 inhibitor alirocumab versus ezetimibe in patients with hypercholesterolemia: results of a 24 week, double-blind, randomized Phase 3 trial. *Int J Cardiol* 2014;**176**:55–61.
50. Koren MJ, Giugliano RP, Raal FJ, Sullivan D, Bolognese M, Langslet G, Civeira F, Somaratne R, Nelson P, Liu T, Scott R, Wasserman SM, Sabatine MS. OSLER Investigators. Efficacy and safety of longer-term administration of evolocumab (AMG 145) in patients with hypercholesterolemia: 52-week results from the open-label study of long-term evaluation against LDL-C (OSLER) randomized trial. *Circulation* 2014;**129**:234–243.

51. Cannon CP, Cariou B, Blom D, McKenney JM, Lorenzato C, Pordy R, Chaudhari U, Colhoun HM. for the ODYSSEY COMBO II Investigators. Efficacy and safety of alirocumab in high cardiovascular risk patients with inadequately controlled hypercholesterolaemia on maximally tolerated doses of statins: the ODYSSEY COMBO II randomized controlled trial. *Eur Heart J* 2015; Feb 16.
52. Nordestgaard BG, Chapman MJ, Humphries SE, Ginsberg HN, Masana L, Descamps OS, Wiklund O, Hegele RA, Raal FJ, Defesche JC, Wiegman A, Santos RD, Watts GF, Parhofer KG, Hovingh GK, Kovanen PT, Boileau C, Averna M, Boren J, Bruckert E, Catapano AL, Kuivenhoven JA, Pajukanta P, Ray K, Stalenhoef AF, Stroes E, Taskinen MR, Tybjaerg-Hansen A. European Atherosclerosis Society Consensus Panel. Familial hypercholesterolaemia is underdiagnosed and undertreated in the general population: guidance for clinicians to prevent coronary heart disease: consensus statement of the European Atherosclerosis Society. *Eur Heart J* 2013;**34**:3478–3490a.
53. Alonso R, Mata N, Castillo S, Fuentes F, Saenz P, Muniz O, Galiana J, Figueras R, Diaz JL, Gomez-Enterria P, Mauri M, Piedecausa M, Irigoyen L, Aguado R, Mata P. Spanish Familial Hypercholesterolaemia Group. Cardiovascular disease in familial hypercholesterolaemia: influence of low-density lipoprotein receptor mutation type and classic risk factors. *Atherosclerosis* 2008;**200**:315–321.
54. Jansen AC, van Aalst-Cohen ES, Tanck MW, Trip MD, Lansberg PJ, Liem AH, van Lennep HW, Sijbrands EJ, Kastelein JJ. The contribution of classical risk factors to cardiovascular disease in familial hypercholesterolaemia: data in 2400 patients. *J Intern Med* 2004;**256**:482–490.
55. Kolansky DM, Cuchel M, Clark BJ, Paridon S, McCrindle BW, Wiegers SE, Araujo L, Vohra Y, Defesche JC, Wilson JM, Rader DJ. Longitudinal evaluation and assessment of cardiovascular disease in patients with homozygous familial hypercholesterolemia. *Am J Cardiol* 2008;**102**:1438–1443.
56. Pijlman AH, Huijgen R, Verhagen SN, Imholz BP, Liem AH, Kastelein JJ, Abbink EJ, Stalenhoef AF, Visseren FL. Evaluation of cholesterol lowering treatment of patients with familial hypercholesterolemia: a large cross-sectional study in The Netherlands. *Atherosclerosis* 2010;**209**:189–194.
57. Robinson JG. Management of familial hypercholesterolemia: a review of the recommendations from the National Lipid Association Expert Panel on Familial Hypercholesterolemia. *J Manag Care Pharm* 2013;**19**:139–149.
58. O'Keefe JH Jr, Cordain L, Harris WH, Moe RM, Vogel R. Optimal low-density lipoprotein is 50 to 70 mg/dl: lower is better and physiologically normal. *J Am Coll Cardiol* 2004;**43**:2142–2146.
59. Cannon CP. IMPROVE-IT trial: a comparison of ezetimibe/simvastatin versus simvastatin monotherapy on cardiovascular outcomes after acute coronary syndromes. *Circulation* 2014;**130**:2105–2126.
60. Toth PP, Harper CR, Jacobson TA. Clinical characterization and molecular mechanisms of statin myopathy. *Expert Rev Cardiovasc Ther* 2008;**6**:955–969.
61. ODYSSEY ALTERNATIVE: efficacy and safety of the proprotein convertase subtilisin/kexin type 9 monoclonal antibody, alirocumab, versus ezetimibe, in patients with statin intolerance as defined by a placebo run-in and statin rechallenge arm. <http://www.abstractsonline.com/pp8/#!/3547/presentation/49567> (28 April 2015).
62. Bays H, Farnier M, Gaudet D, Weiss R, Lima Ruiz J, Watts GF, Gouni-Berthold I, Robinson JG, Jones PH, Severance R, Averna M, Steinhagen-Thiessen E, Colhoun HM, Zhao J, Du Y, Hanotin C, Donahue S. Efficacy and safety of combining alirocumab with atorvastatin or rosuvastatin versus statin intensification or adding ezetimibe in high cardiovascular risk patients: ODYSSEY OPTIONS I and II. *Circulation* 2014;**130**:2118–2119.
63. Efficacy and safety of alirocumab in patients with heterozygous familial hypercholesterolaemia not adequately controlled with current lipid-lowering therapy: results of ODYSSEY FH I and FH II studies. <http://www.escardio.org/congresses/esc-2014/congress-reports/Pages/707-3-Hotline3-ODYSSEY-COMBO-FH.aspx#presenter2> (28 April 2015).
64. Etemad LR, McCollam PL. Total first-year costs of acute coronary syndrome in a managed care setting. *J Manag Care Pharm* 2005;**11**:300–306.
65. Conly J, Clement F, Tonelli M, Hemmelgarn B, Klarenbach S, Lloyd A, McAlister FA, Huserau D, Wiebe N, Au F, Manns B. Alberta Kidney Disease Network. Cost-effectiveness of the use of low- and high-potency statins in people at low cardiovascular risk. *CMAJ* 2011;**183**:E1180–8.
66. Moriarty PM. LDL-apheresis therapy. *Curr Treat Options Cardiovasc Med* 2006;**8**:282–288.
67. Nherera L, Calvert NW, Demott K, Humphries SE, Neil HA, Minhas R, Thorogood M. Cost-effectiveness analysis of the use of a high-intensity statin compared to a low-intensity statin in the management of patients with familial hypercholesterolaemia. *Curr Med Res Opin* 2010;**26**:529–536.
68. Robinson JG, Farnier M, Krempf M, Bergeron J, Luc G, Averna M, Stroes ES, Langslet G, Raal FJ, El Shahawy M, Koren MJ, Lepor NE, Lorenzato C, Pordy R, Chaudhari U, Kastelein JJ, ODYSSEY LONG TERM Investigators. Efficacy and safety of alirocumab in reducing lipids and cardiovascular events. *N Engl J Med* 2015;**372**:1489–1499.
69. Urban D, Pöss J, Böhm M, Laufs U. Targeting the proprotein convertase subtilisin/kexin type 9 for the treatment of dyslipidemia and atherosclerosis. *J Am Coll Cardiol* 2013;**62**:1401–1408.
70. Dadu RT, Ballantyne CM. Lipid lowering with PCSK9 inhibitors. *Nat Rev Cardiol* 2014;**11**:563–575.
71. Koren MJ, Roth EM, McKenney JM, Gipe D, Hanotin C, Ferrand AC, Wu R, Dufour R. Safety and efficacy of alirocumab 150 mg every 2 weeks, a fully human proprotein convertase subtilisin/kexin type 9 monoclonal antibody: A Phase II pooled analysis. *Postgrad Med* 2015;**127**:125–132.
72. Gouni-Berthold I, Berthold HK. PCSK9 antibodies for the treatment of hypercholesterolemia. *Nutrients* 2014;**6**:5517–5533.